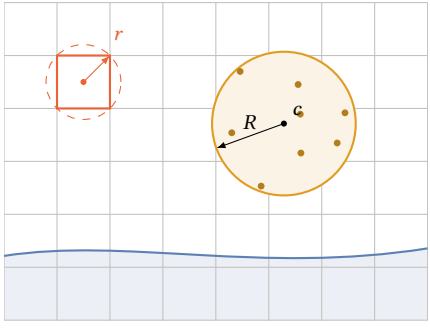
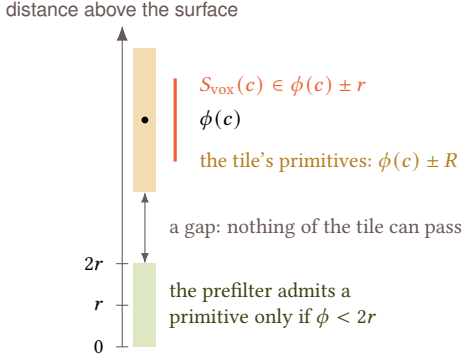


a voxel: S_{vox} is ϕ at its centre, within r

a tile: 8 primitives within R of c



the gap exists if $\phi(c) - R \geq 2r$, and $S_{\text{vox}}(c) \geq R + 3r$ guarantees it: so a tile is rejected only when none of its primitives can be a contact